Amber

An app centered on memories and social connections

1. The principle behind the app

Amber is an Android app designed for people with mild to moderate memory disorders, as well as their loved ones and carers. Its aim is to use personal memories as a means of cognitive stimulation and interaction, whilst offering a simple experience tailored to the needs of those affected

The app is based on two complementary sections, which are as follows.

- **Relatives, family members and carers** can gradually build up a library of memories. They can add photographs, specify the people, dates and places associated with them, write descriptions and record short voice messages linked to these memories
- **The person receiving care** accesses this content via a deliberately simplified interface: limited text and large buttons, minimizing distractions and usability issues as much as possible

Our aim is not simply to create a digital photo album. Amber seeks to make memories **interactive**, even when there is no direct contact with loved ones. Using the information provided by them, the app can suggest questions or conversation starters based on a photograph – for example, about a person, an event or a place. Eventually, these exchanges may also take the form of a more interactive conversation, allowing the user to recount their memories verbally and gradually build narratives based on their responses.

The app is thus designed to act as an intermediary **between the user and their loved ones**. Loved ones contribute media to the app; the user can then rediscover these at their own pace and use them as a starting point for reminiscing, sharing stories and engaging in conversation.

Certain features under consideration would also make it possible to tailor the experience further to each individual: choosing which memories or people the user wishes to see, the option to flag content as not to their liking, grouping memories by person or by story, linking memories to places, or even generating timelines and narratives from multiple memories.

Finally, it was deemed necessary for some of the content to remain available **offline**, so that the app can be used even when internet access is limited.

2. Why Amber can be useful

Amber is not intended to treat or cure dementia. Rather, the app is designed as a **complementary tool to stimulate and maintain social connections**, drawing inspiration in particular from two approaches: **reminiscence therapy** and **cognitive stimulation**.

Reminiscence therapy

Reminiscence therapy is based on evoking personal memories, often with the help of aids such as photographs, sounds or stories.

Amber adapts this principle to a digital and personalized environment. The photographs used are not generic images: they come directly from the person's own life and that of their family. As described in the previous section, they may be accompanied by names, dates, places, descriptions and associated voices.

The aim is therefore to provide elements likely to facilitate the recall of memories.

Cognitive stimulation

Amber also draws inspiration from cognitive stimulation by encouraging interaction around the memories displayed.

Rather than simply displaying a photograph, the app can use its **context** to pose a question or invite the user to share a story, for example:

'Do you recognize this person?'

'What were you doing that day?'

Ultimately, the aim is to facilitate a genuine exchange around this content, particularly through speech recognition and a conversational interface.

However, the aim is not to turn the experience into a memory test. The app must avoid, as far as possible, putting the user in a situation where they might fail or feel frustrated. The interactions, if implemented, are intended first and foremost as conversation starters.

Encouraging interaction with loved ones

Amber also aims to encourage interaction between people with memory disorders and those around them.

A photograph or a voice message can serve as a starting point for a conversation during a visit or simply help the person to reconnect with the voice and memories associated with a loved one who is not physically present.

It is also planned that those affected will be able to add their own photos to the app, so that they can actively contribute to their collection of memories and view them at any time. This could then lead to a discussion with loved ones about these memories.

Amber therefore seeks not only to **display memories**, but to create a space where they can **be brought to life, recounted and shared**.

3. What we aim to offer

Through Amber, we therefore aim to provide a tool that can:

- Facilitate access to personal and meaningful memories
- Encourage the recall of memories, conversation and cognitive activity centered on these memories
- Foster interactions between the person concerned and their loved ones, even from a distance
- Enable families to easily contribute to a shared library of memories
- Offer a simple and personalized experience, tailored to each person's abilities and preferences
- Gradually enable the creation and preservation of stories based on memories recounted by the person themselves

These objectives currently form the main assumptions underpinning the project. A key part of our approach is precisely to test these against the experiences of professionals and families to understand **what can genuinely be beneficial, what needs to be adapted and what limitations must be taken into account**.

4. The profiles we are looking for

In order to develop Amber in a responsible and meaningful way, we would like to gather the views of people who are directly affected by memory-impairing conditions, whether from a professional or personal (e.g. family-related) perspective.

We are primarily looking for two types of participants.

Healthcare professionals and specialists in memory disorders

We would like to speak with professionals working in the fields of neurology, geriatrics, neuropsychology or, more generally, in the support and care of people with neurodegenerative or memory-impairing conditions.

The aim is to draw on their expertise to better understand:

- Whether the app's general approach seems relevant to them
- In which situations this type of tool might be useful or, conversely, unsuitable
- What limitations or precautions should be taken into account
- How to use memories and questions without causing unnecessary frustration, anxiety or confusion
- What types of interactions would be most appropriate depending on the stage of the disease
- Which features should be prioritized or avoided for the patient's well-being
- How to better adapt the app to the real needs of people with dementia and their carers

We are particularly interested in receiving a **critical** assessment of the project. Comments regarding the limitations of our approach, potential risks or necessary changes are therefore just as important as positive feedback.

Families and carers supporting someone with memory problems

We are also looking for **families or carers supporting someone with mild to moderate memory problems** who would be willing to trial Amber for a short period.

The idea is that one or more carers would add a few photographs, descriptions or voice messages to the app, so that the person concerned can use the interface designed for them.

We would then like to gather feedback from the person and their careers, particularly regarding:

- How easy the app is to use
- Understanding the interface – what should be kept, removed, changed or added
- The interest sparked by the photographs and memories presented
- Reactions to questions or prompts to share stories associated with the memories
- Features that are appreciated or deemed unnecessary
- Any situations that may have caused confusion or frustration
- Features that are missing and would make the app truly useful in everyday life

These tests are not intended to measure medical or therapeutic effectiveness. Their primary aim is to understand **how the app is experienced in a real-life situation**, so that we can modify its functionality, its interface and the way in which memories are presented.

Every precaution will be taken to ensure the testing period runs smoothly and without a hitch, with prior advice from healthcare professionals on how to proceed.